

GRADE 9 MATHEMATICS · ALGEBRA

Linear Equations & Inequalities

THE BIG IDEA

Every equation is a balance scale. If you do something to one side, you must do to the other.

SIX STEPS TO SOLVE

1. **Distribute** — expand any brackets
2. **Combine like terms** on the same side

ctions es

e. Whatever you do to one
her — until **x** stands alone.



3. **Move variables** to one side

4. **Move constants** to the other side

5. **Divide** to isolate the variable

6. **Verify** by substituting back

Inequalities: solve exactly the same way — **except** negative.

WORKED EXAMPLE

Solve: $4(x - 3) = 2x + 6$

Distribute: $4x - 12 = 2x + 6$

Move variables: $2x - 12 = 6$

Move constants: $2x = 18$

Divide: $x = 9$

Verify: $4(6) = 24$ $2(9)+6 = 24$ ✓

Do not flip the sign when you multiply or divide by a

 WATCH OUT FOR

MISTAKE

Distributing to only the first term

MISTAKE

Not verifying the answer

MISTAKE

Flipping the sign when adding or subtracting

 SELF-CHECK

Can you solve $2(3x - 1) = -(x + 4)$ and verify

FIX

Multiply **every** term inside the brackets

FIX

Always substitute back into the original equation

FIX

Only flip when multiplying or dividing by a
negative

? Can you solve $-5x < 20$ and graph it?